

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, 2D image slices are acquired in one pre-defined direction one after the other, and the final 3D volume is created by stacking the 2D slices. For instance, in axial acquisitions 2D slices are acquired in the transverse plane, in sagittal acquisitions in the sagittal plane, and in coronal acquisitions in the coronal plane. The voxel sizes of 2D acquisitions are given based on the acquisition direction, as in-plane and through-plane voxel sizes, where the first is represented with two values (pixel size in the imaging plane) and the latter corresponds to a single value (length of the voxel in the third dimension). Recall that the field-of-view is the real-world size of the entire image, i.e. image size $\times$ voxel size. Based on this please compute the following.
 - (a) A *sagittal* acquisition with voxel size in-plane $0.5 \times 0.5 \text{ mm}^2$ with 384 pixels in both directions and voxel size through-plane 4.0 mm with 60 pixels is acquired. What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

Solution: Answer: $192 \times 240 \times 192 \text{ mm}^3$.

The sagittal plane contains the head-to-toe and the anterior-to-posterior directions, so both in-plane directions give $384 \times 0.5 = 192 \text{ mm}$. The through-plane direction of a sagittal acquisition is left-to-right: $60 \times 4.0 = 240 \text{ mm}$.

- (b) One year later the same patient is scanned again, this time with an *axial* acquisition with voxel size in-plane $0.8 \times 0.8 \text{ mm}^2$ with a field-of-view of 192 mm in both directions, and voxel size through-plane of 2.0 mm with a field-of-view of 240 mm. What are the number of voxels in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

Solution: Answer: $120 \times 240 \times 240$ voxels.

The through-plane direction of an axial acquisition is head-to-toe: $240/2.0 = 120$. Both in-plane directions give $192/0.8 = 240$.

- (c) You want to compare a volumetric measurement of the same anatomical structure in the two images. Following the guideline given in the lecture, you resample the second image (b) onto the voxel grid of the first image (a). What are the resize factors along the head-to-toe, left-to-right and anterior-to-posterior directions, and what is the resulting image size? Why do you match the voxel size and not the image size?

Solution: Answer: resize factors $4.0 \times 0.2 \times 1.6$, resulting image size $480 \times 48 \times 384$ voxels. The resize factor is the ratio of the old to the new voxel size, per direction: head-to-toe $2.0/0.5 = 4.0$, left-to-right $0.8/4.0 = 0.2$, anterior-to-posterior $0.8/0.5 = 1.6$. Applying these to (b): $120 \cdot 4.0 = 480$, $240 \cdot 0.2 = 48$, $240 \cdot 1.6 = 384$. The field-of-view is unchanged ($240 \times 192 \times 192 \text{ mm}^3$), as it must be.

Measurements taken in an image correspond to real-world quantities only through the voxel size, so the voxel size is what has to be matched. Matching the image sizes instead would silently change the effective voxel size (and here also the aspect ratio), and the two measurements would no longer be comparable.

2. Many noise suppression methods can be written as a weighted average $\hat{I}(x) = \sum_y w(x, y) J(y)$ of the noisy image J , with $w(x, y) \geq 0$ and $\sum_y w(x, y) = 1$. Which of the two weight definitions below is the one used by the *non-local means* filter? $N(x)$ denotes the image patch centred at x , $Z(x)$ is a normalisation constant and h the degree of filtering. Please circle the correct choice.

$$w(x, y) = \frac{1}{Z(x)} \exp \left(-\frac{\|J(x) - J(y)\|_2^2}{h^2} \right)$$

$$w(x, y) = \frac{1}{Z(x)} \exp \left(-\frac{\|J(N(x)) - J(N(y))\|_2^2}{h^2} \right)$$

Solution: Answer: Right one.

Non-local means compares the whole geometric configuration of a *patch*, not a single grey value. The left expression compares single (noisy) intensities and is the Yaroslavsky / bilateral type of neighbourhood filter, which is much less robust: two pixels can share an intensity by chance while sitting in completely different structures.

3. Deformable intensity-based registration is usually written in one of the two equivalent forms below. The first is derived from a probabilistic model, the second is energy-based; T_θ is the transformation with parameters θ , I the target and J the moving image.

$$\max_{\theta} \log p(I | T_\theta \circ J) + \log p(T_\theta) \qquad \min_{\theta} d(I, T_\theta \circ J) + \lambda r(T_\theta^{-1})$$

- (a) Please indicate which term corresponds to which term in the two formulations.

Solution: Answer: $\log p(I | T_\theta \circ J) \iff d(I, T_\theta \circ J)$ and $\log p(T_\theta) \iff \lambda r(T_\theta^{-1})$.

- (b) Deformable registration is an ill-posed problem. Which of the two terms addresses this, and how?

Solution: Answer: the regularisation term $\lambda r(T_\theta^{-1})$, equivalently the prior $p(T_\theta)$. Many very different transformations explain the intensities about equally well, so the data term alone does not determine a unique solution. The regulariser expresses a preference

among them — e.g. ℓ_2 regularisation of ∇u prefers smoothly varying displacement fields, ℓ_1 prefers piecewise constant ones, and elastic / hyperelastic energies prefer small or locally rigid deformations.

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.
- (a) True **FALSE** In the discrete setting, the *forward* transformation is preferred over the backward (inverse) transformation for resampling an image, because it guarantees that every pixel of the output grid is assigned a value.
 - (b) **TRUE** False Nearest-neighbour interpolation is the appropriate choice for transforming a label (segmentation) map, because it does not introduce new values.
 - (c) True **FALSE** Bilinear interpolation produces a function that is both continuous and differentiable.
 - (d) True **FALSE** In 3D, Keys' cubic convolution (tricubic) interpolation uses only the 8 corner voxels of the cell that contains the query point.

Solution: (a) FALSE — it is exactly the other way around. Under a forward mapping the transformed input pixels do not land on output grid points, two input pixels may land on the same output pixel, and some output pixels receive nothing at all (holes). The backward mapping asks, for each output pixel, where it comes from in the input image, and interpolates there; every output pixel therefore gets a value.

(b) TRUE — nearest neighbour only copies existing values, so no labels that do not exist can appear. Bilinear or bicubic interpolation would create spurious intermediate label values at the boundaries.

(c) FALSE — bilinear interpolation yields a piecewise-linear function: continuous, but not differentiable at the grid lines. Bicubic interpolation is the one that is both continuous and differentiable.

(d) FALSE — the 8 corner voxels are what *trilinear* interpolation uses. Keys' kernel has support $[-2, 2]$, covering 4 samples per dimension, i.e. $4 \times 4 = 16$ pixels in 2D and $4 \times 4 \times 4 = 64$ voxels in 3D.

5. Mixture models and Expectation-Maximization (EM) segmentation are essential building blocks for many segmentation algorithms. Consider a multi-modal image in which every pixel carries *two* intensity values (for instance a co-registered T1-weighted and T2-weighted pair). How many parameters does a Gaussian Mixture Model (GMM) with 3 components have? Count, for each component, the mean vector, the full covariance matrix and the mixture weight.

Solution: Answer: $3 \times (2 + 3 + 1) = 18$.

Per component: a mean vector in $\mathbb{R}^2$ gives 2 parameters; a full 2×2 covariance matrix is symmetric and therefore has 3 free parameters (σ_1^2 , σ_2^2 and one covariance); the mixture weight gives 1. That is 6 parameters per component and 18 in total.

6. An Active Shape Model (ASM) is fitted to an image iteratively. Once the model has been initialised and candidate points have been found by sampling image features along the surface normals of the model points, each iteration updates the model in two parts. Please briefly describe what these two parts do, and state what keeps the fitted shape anatomically plausible. You do not need to write down the equations.

Solution: Answer: *Pose estimation* computes a global transformation — e.g. a similarity transform, i.e. translation, rotation and isotropic scaling — that best aligns the model points with the candidate points found along the normals, for instance by minimising the mean Euclidean distance between the two point sets (ICP). *Shape estimation* then removes the pose and computes the shape parameters b of the PCA (point distribution) model in closed form, which deforms the mean shape along its modes of variation. The two are iterated until convergence.

Plausibility is enforced by the model itself: only the leading principal modes are kept (e.g. as many as needed to explain about 90% of the variance), and each shape parameter is constrained to $|b_i| < 3\sigma_i$, so the fit can only produce shapes that are typical of the training population.

7. Thin-plate splines (TPS) and free-form deformations (FFD) with cubic B-splines are both interpolation-based models for non-linear transformations. Which of the following statements on their similarities and differences is **inaccurate**?
- ☐ (A) Both models place displacement vectors on a set of control points and interpolate in between with a smooth basis function, so that the resulting transformation is smooth for any value of the parameters.
 - ☐ (B) The influence of a single control point is global in TPS, whereas for FFD with cubic B-splines it is local, because the B-spline basis functions have finite support.
 - ☐ (C) In FFD, increasing the number of control points, i.e. using a finer control grid, results in smoother transformations.
 - ☐ (D) For a given set of N control points in 3D, both models are parameterised by the $3N$ coefficients of the displacement vectors at the control points.

Solution: Answer: C.

It is the other way around: a finer control grid gives the model more degrees of freedom locally and therefore produces *less* smooth transformations. This is precisely what makes multi-resolution registration possible — start with few control points for a coarse, very smooth alignment and progressively refine the grid.

8. Multi-atlas and patch-based methods were the state of the art in segmentation for a long time. Which of the following statements about them is **incorrect**?
- ☐ (A) In multi-atlas segmentation, every atlas is registered to the unseen image, each atlas produces its own propagated segmentation, and these are combined in a decision fusion step such as majority voting or STAPLE.

- ☐ (B) STAPLE estimates the most probable underlying true segmentation and, simultaneously, the sensitivity and specificity of each of the input segmentations; it is solved iteratively with an Expectation-Maximization algorithm.
- ☐ (C) Selecting only the atlases that are similar to the unseen image before fusing performs worse than naively fusing all available atlases, because discarding atlases throws away information.
- ☐ (D) Patch-based label fusion requires only a coarse, rigid or affine registration, because correspondence is re-established locally by comparing patches inside a search window.

Solution: Answer: C.

Atlas selection followed by fusion *outperforms* naive fusion of all atlases: atlases that are dissimilar to the target contribute mostly registration errors, and removing them improves the fused result. The same reasoning underlies weighted voting, where atlases are weighted by how well they match the target rather than being discarded outright.

9. Convolutional neural networks and Vision Transformers (ViT) build very different assumptions into the model. Which of the following statements is **incorrect**?
- ☐ (A) In CNNs, locality and the two-dimensional neighbourhood structure of the image act as strong inductive biases for vision tasks.
 - ☐ (B) In a ViT, self-attention is global and only the MLP layers act locally, so that splitting the image into patches is essentially the only two-dimensional inductive bias built into the model.
 - ☐ (C) Because a ViT builds in fewer assumptions about images, it typically requires *less* training data than a comparable CNN to reach the same accuracy.
 - ☐ (D) Without positional encodings, a transformer encoder is equivariant to permutations of its input tokens, which is why positional information has to be added to the token embeddings.

Solution: Answer: C.

Fewer built-in assumptions means the geometric properties of images that a CNN gets for free have to be learned from the data instead. ViTs therefore generally require *more* training data than a comparable CNN, not less.

10. You are training a U-Net to segment a small lesion that occupies far less than 1% of the image. Which of the following statements is correct?
- ☐ (A) With pixel-wise cross-entropy, the loss contributed by the lesion is typically much smaller than the loss contributed by the rest of the image, so the lesion can be dominated during training.
 - ☐ (B) The Sørensen-Dice coefficient computed on the binary label sets is differentiable and can therefore be used directly as a training loss.
 - ☐ (C) The Dice loss makes the contribution of a class proportional to its size, which is why it helps for small structures.

- (D) The skip connections of the U-Net remove any need to reconsider the loss function, because they restore the resolution lost in the encoder.
- (E) Multi-class cross-entropy cannot be used when more than two classes are present.

Solution: Answer: A.

The cross-entropy loss decomposes into a sum over pixels, $\mathcal{L} = \mathcal{L}_{\Omega_{small}} + \mathcal{L}_{\overline{\Omega_{small}}}$, and since $|\Omega_{small}| \ll |\overline{\Omega_{small}}|$ the first term is normally negligible. (B) is wrong because the Dice coefficient as a set measure is *not* differentiable, which is why the soft Dice loss is used instead. (C) is wrong in the opposite direction: the soft Dice loss gives each class a contribution of comparable magnitude *regardless* of its size, which is exactly why it helps here. (D) confuses resolution with class imbalance — skip connections address the former, not the latter.

11. Consider the encoder of a segmentation network that takes grey-scale input images of size 128×128 with one input channel and has the following architecture:

- **Conv1:** 32 kernels with kernel size $K = 3 \times 3$, stride $S = 1 \times 1$ in each direction, and “same” padding $P = 1$.
- **Pool1:** Max pooling with $K = 2 \times 2$ and $S = 2$.
- **Conv2:** 64 kernels with $K = 3 \times 3$, $S = 1$ and “same” padding $P = 1$.
- **Pool2:** Max pooling with $K = 2 \times 2$ and $S = 2$.
- **Conv3:** 128 kernels with $K = 3 \times 3$, $S = 1$ and “same” padding $P = 1$.
- **Pool3:** Max pooling with $K = 2 \times 2$ and $S = 2$.

Circle True or False:

- (a) **TRUE** False The global receptive field of a neuron in Pool3 is 22×22 .
- (b) True **FALSE** The global stride of a neuron in Pool3 is 3×3 , one for each pooling layer.
- (c) **TRUE** False The total number of convolutional kernel weights (excluding the biases) in Conv1, Pool1, Conv2, Pool2, Conv3 and Pool3 is $3 \cdot 3 \cdot 1 \cdot 32 + 3 \cdot 3 \cdot 32 \cdot 64 + 3 \cdot 3 \cdot 64 \cdot 128 = 92\,448$.
- (d) True **FALSE** The output of Pool3 has a spatial size of 16×16 , and each of its neurons therefore sees the entire 128×128 input image.

Solution: (a) TRUE — track the receptive field R and the jump (global stride) j layer by layer, starting from $R = 1$, $j = 1$, using $R \leftarrow R + (K - 1) \cdot j$ and $j \leftarrow j \cdot S$: Conv1 $\rightarrow R = 3$, $j = 1$; Pool1 $\rightarrow R = 4$, $j = 2$; Conv2 $\rightarrow R = 8$, $j = 2$; Pool2 $\rightarrow R = 10$, $j = 4$; Conv3 $\rightarrow R = 18$, $j = 4$; Pool3 $\rightarrow R = 22$, $j = 8$.

(b) FALSE — strides compound multiplicatively, they do not add: three poolings of stride 2 give a global stride of $2 \cdot 2 \cdot 2 = 8$, i.e. 8×8 .

(c) TRUE — $288 + 18\,432 + 73\,728 = 92\,448$. Pooling layers have no learnable parameters at all.

(d) FALSE — the spatial size is indeed $128 \rightarrow 64 \rightarrow 32 \rightarrow 16$, since “same” convolutions do not change the size and each pooling halves it. But the output size and the receptive field are different things: by (a) a neuron in Pool3 only sees a 22×22 window of the input, far less than the whole image. This is precisely why deeper networks, or skip connections combined with more downsampling, are needed to gather global context.

12. Which of the following statements on uncertainty estimation, domain shift and learning from unlabelled examples are correct (multiple answers are possible)?
- ☐ (A) Standard, deterministic training and inference amount to approximating $p(\theta \mid D, M)$ by a Dirac delta at θ^* and $p(y \mid x, \theta, M)$ by a Dirac delta at $f(x; \theta)$, so that the resulting predictive distribution is a point mass carrying no uncertainty at all.
 - ☐ (B) Test-time adaptation still requires labelled samples from the target domain, but fewer of them than fine-tuning does.
 - ☐ (C) Self-training with pseudo-labels works well only if the predictions of the initial model are reasonably good; with poor initial estimates it can diverge quickly.
 - ☐ (D) In the mean-teacher model the teacher is trained by minimising the supervised loss on the labelled examples, while the student is the exponential moving average of the teacher.
 - ☐ (E) Domain generalisation by extreme data augmentation trains only in the source domain, but the improvement it brings may be limited to the kinds of variation that the augmentations actually cover.

Solution: Answer: (A), (C) and (E) are correct.

(B) is wrong: test-time adaptation uses *no* labels and no training set at the target domain — it adapts a subset of the parameters at inference time by minimising a label-free cost (prediction entropy, an auto-encoder reconstruction loss, or the agreement of the output with a source-trained segmentation prior). It is fine-tuning that needs a few labelled target samples.

(D) has the two networks swapped: the student is the one trained on the supervised plus consistency loss, and the teacher is the exponential moving average of the student, $\phi^i = \alpha\phi^{i-1} + (1 - \alpha)\theta^i$.